

BRASOV-GHIMBAV, Romania

WMO: 153000

Lat: 45.6958N

Lon: 25.5261E

Elev: 535

StdP: 95.06

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.9	-18.5	0.8	-16.6	-15.7	1.0	-13.1	13.0	3.0	10.2	2.0	0.8	230	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.5	30.1	19.4	28.5	19.0	27.1	18.6	20.9	27.6	20.1	26.5	19.3	25.4	3.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.4	23.3	17.9	13.7	22.2	17.2	13.1	21.7	62.8	27.8	59.9	26.5	57.5	25.5	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.7	7.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-3.7	0.0	3.7	8.8	14.3	18.2	19.0	19.4	14.7	9.4	4.2	-0.8	(6)
(7)		DBStd	8.89	5.40	5.21	4.52	4.87	3.10	2.80	2.64	2.62	4.02	3.91	4.81	4.58	(7)
(8)		HDD10.0	1569	425	279	197	79	6	1	0	0	7	58	182	334	(8)
(9)		HDD18.3	3555	683	512	452	287	126	36	25	20	119	277	424	592	(9)
(10)		CDD10.0	1199	0	1	3	42	141	247	279	291	147	40	8	0	(10)
(11)		CDD18.3	142	0	0	0	0	3	32	46	53	9	0	0	0	(11)
(12)		CDH23.3	1645	0	0	0	13	78	337	442	604	164	5	0	0	(12)
(13)		CDH26.7	411	0	0	0	0	7	68	113	185	38	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.1	2.5	3.2	3.1	2.5	2.5	2.6	2.2	2.5	2.3	2.3	2.3	(14)
(15)	Precipitation	PrecAvg	594	27	24	28	46	73	90	93	72	46	35	32	28	(15)
(16)		PrecMax	803	69	44	76	100	169	160	184	132	123	136	76	58	(16)
(17)		PrecMin	383	5	0	2	20	15	31	21	15	3	4	5	1	(17)
(18)		PrecStd	100	18	11	21	19	36	34	40	32	31	25	17	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	16.2	19.8	24.9	27.6	30.1	31.3	32.4	30.3	23.4	19.1	12.0	(19)
(20)			MCWB	6.7	11.0	10.8	14.3	16.0	19.6	19.7	20.7	18.5	14.7	12.2	8.4	(20)
(21)		2%	DB	6.4	11.8	16.3	23.0	25.7	28.6	29.5	30.4	27.2	21.5	16.2	9.3	(21)
(22)			MCWB	4.1	7.6	9.3	13.9	16.0	19.6	19.7	19.4	17.4	13.7	11.3	6.9	(22)
(23)		5%	DB	4.1	9.5	13.5	20.2	23.6	27.0	27.8	28.6	24.8	19.1	13.7	7.0	(23)
(24)			MCWB	2.6	6.3	7.9	12.4	15.3	19.0	19.3	18.8	16.5	12.9	9.9	5.1	(24)
(25)		10%	DB	2.6	7.1	10.8	17.6	21.7	25.3	25.8	26.9	22.6	16.4	11.6	4.8	(25)
(26)			MCWB	1.6	4.8	6.3	11.2	14.6	18.5	18.8	18.6	15.8	11.9	8.7	3.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	10.8	11.5	15.3	18.3	22.1	21.6	21.7	19.5	15.9	12.7	8.9	(27)
(28)			MCDWB	9.6	15.6	19.1	22.8	24.4	28.3	27.9	29.6	27.7	21.1	17.6	10.8	(28)
(29)		2%	WB	4.4	8.0	9.6	14.2	16.9	20.6	20.7	20.8	18.2	14.5	11.6	7.3	(29)
(30)			MCDWB	6.1	11.4	15.8	21.5	23.1	26.6	27.0	28.2	24.7	19.7	15.9	9.4	(30)
(31)		5%	WB	2.9	6.6	8.3	12.9	15.9	19.7	20.0	19.8	17.1	13.5	10.5	5.3	(31)
(32)			MCDWB	4.1	9.3	13.1	19.7	22.0	25.7	26.0	26.6	23.7	18.0	13.9	7.1	(32)
(33)		10%	WB	1.7	4.9	6.9	11.7	15.1	18.9	19.2	19.0	16.3	12.3	8.9	3.7	(33)
(34)			MCDWB	2.5	6.8	10.2	16.8	20.3	24.6	24.7	25.4	21.9	16.0	11.4	4.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	9.1	9.8	11.6	11.7	11.9	11.9	13.5	11.8	11.5	8.7	6.2	(35)
(36)			MCDBR	7.9	12.3	15.2	16.7	15.9	14.6	15.6	17.0	16.8	17.1	12.7	9.3	(36)
(37)			MCWBR	5.7	8.2	8.8	8.4	7.5	6.7	6.9	6.9	7.5	9.6	7.9	6.9	(37)
(38)		5% WB	MCDWB	7.5	12.2	14.1	15.2	14.0	13.7	13.7	15.0	15.1	15.4	12.1	9.0	(38)
(39)			MCWBR	5.6	8.2	8.3	7.8	6.9	6.6	6.3	6.6	7.0	8.9	7.5	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.327	0.369	0.426	0.408	0.408	0.424	0.436	0.392	0.355	0.329	0.300	(40)	
(41)		taud	2.341	2.276	2.242	2.136	2.255	2.289	2.248	2.211	2.307	2.402	2.423	2.384	(41)	
(42)		Ebn at Noon	800	842	853	831	861	861	840	813	820	803	760	758	(42)	
(43)		Edh at Noon	76	99	119	144	133	129	133	133	111	87	71	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.38	2.20	3.28	4.29	5.24	5.66	5.73	5.21	3.80	2.62	1.53	1.11	(44)	
(45)		RadStd	0.15	0.27	0.35	0.46	0.51	0.51	0.49	0.45	0.38	0.33	0.16	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)	+0.60	N/A	N/A	+0.69	N/A	N/A	-132	-178	+76	+25

Nomenclature: See separate page